

# GI Neuroendocrine Tumors

## Gastric Neuroendocrine Tumors

| Category         | Differentiation | Grade     | Mitotic Rate | Ki-67 Index |
|------------------|-----------------|-----------|--------------|-------------|
| NET - Grade 1    | Well-diff       | Low       | <2           | <3%         |
| NET - Grade 2    | Well-diff       | In-termed | 2-20         | 3-20%       |
| NET - Grade 3    | Well-diff       | High      | >20          | >20%        |
| NEC - Large Cell | Poorly          | High      | >20          | >20%        |
| NEC - Small Cell | Poorly          | High      | >20          | >20%        |
| MiNEN            | Variable        | Variable  | Variable     | Variable    |

## Mixed Neuroendocrine - Nonneuroendocrine Neoplasms

### Somatostatin Receptor (SSTR) PET for Staging

[Reference Here](#)

Frequently performed with  $^{68}\text{Ga}$ -DOTATATE Most useful for poorly-differentiated or high-grade tumors Less helpful for small tumors

## Gastric Neuroendocrine Tumors

Type 1 Multifocal Tumors - Driven by atrophic gastritis (70–80% of gastric NETs):

- Atrophic gastritis  $\rightarrow$   $\downarrow$  gastric acid ( $\uparrow$ pH)  $\rightarrow$   $\uparrow$  gastrin from G cells
- Gastrin is trophic for enterchromaffin-like (ECL) cells  $\rightarrow$  Neuroendocrine tumors
- Small ( $<1$  cm), multifocal, well-differentiated, indolent

Type 2 Multifocal Tumors - Driven by Zollinger-Ellison Syndrome (rare)

- Gastrinoma produces high levels of gastrin and low gastric pH
- Generally have accompanying ulcer disease

Type 3 Sporadic Tumors - Autonomous growth (normal pH and normal gastrin)

Highest metastatic potential; most aggressive subtype

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- Small ( $<1$  cm), multifocal, well-differentiated, indolent
- Small tumors  $<1$ cm: Endoscopic resection
- Tumors 1-2cm: Endoscopic mucosal resection (EMR) or Endoscopic Submucosal Dissection (ESD)
- High risk tumors ( $>2$ cm, Node<sup>+</sup>, Grade 3, muscle invasion): Surgical resection

## Gastric Neuroendocrine Tumors

Type 2 Multifocal Tumors - Driven by Zollinger-Ellison Syndrome (rare)

- Gastrinoma produces high levels of gastrin and low gastric pH
- Generally have accompanying ulcer disease
- Localization of gastrinoma (and resection) critical to treatment

## Gastric Neuroendocrine Tumors

Type 3 Sporadic Tumors - Autonomous growth (normal pH and normal gastrin)

Highest metastatic potential; most aggressive subtype

- Small tumors <1cm: Endoscopic resection
- High risk tumors: Surgical resection with node dissection

## Mixed Neuroendocrine / Nonneuroendocrine Neoplasms

- Tumors showing neuroendocrine *and* nonneuroendocrine differentiation
- Tend to be aggressive with liver and nodal metastasis
- Origin from a single stem cell that then gives rise to two differentiation paths
- Small tumors treated with surgery
- Larger tumors treated with chemotherapy
- Also known as Mixed Adenoneuroendocrine Carcinoma (MANEC)

(jacob721?)

## Lutathera Lu<sup>177</sup>-dotatate

- Radioactive-tagged dotatate (artificial somatostatin analogue)
- Indicated for treatment of somatostatin receptor (SSTR)-positive tumors
- Gastroenterohepatic neuroendocrine tumors of foregut, midgut, and hindgut
- Dosed IV every 8 weeks x 4 doses, followed by long-acting octreotide every 4 weeks

## NETTER-1 clinical trial

Eligibility criteria:

- Neuroendocrine tumors
- Ki-67 index ≤ 20%
- OctreoScan positive in all lesions
- Disease progression on standard-dose octreotide

Randomized:

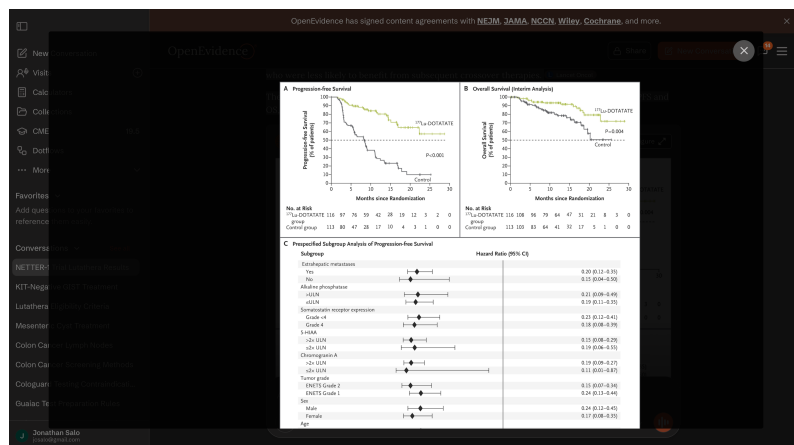
- <sup>177</sup>Lu-dotatate q8 weeks x4 + octreotide LAR 30mg IM  
OR
- Octreotide LAR 60mg IM every 4 weeks

|                                  | <sup>177</sup> Lu-dotatate | Control |
|----------------------------------|----------------------------|---------|
| Progression-free survival @ 20mo | 65%                        | 11%     |
| Median PFS                       | (not reached)              | 8.4mo   |
| Objective Response Rate          | 18%                        | 3%      |
| Overall Survival Median          | 48mo                       | 33mo    |

Survival difference was likely diluted by crossovers at progression

More pronounced survival benefit for patients with KPS 60-80

## NETTER-1



## Lutathera indications for GI/Pancreatic tumors

- Well differentiated (Grade 1/2) tumors
  - Progression on octreotide LAR or lanreotide
  - Significant tumor burden and Ki-67 index  $\geq 10\%$  (first line)
- Well differentiated Grade 3 tumors
  - Favorable biology (Ki-67 10%-55%) - supported by NETTER-2 clinical trial

## Netter-2 Trial - Grade 2/3 Neuroendocrine tumors

226 Patients with higher grade 2 (Ki67 10% and 20%) and grade 3 (Ki67  $>20\%$  and 55%), somatostatin receptor-positive gastroenteropancreatic neuroendocrine tumours

Treated with 4 cycles of Lutathera → Octreotide 30mg IM q4 wk

vs

Octreotide 60mg IM q4wk

Impact of Liver Tumour Burden, Alkaline Phosphatase Elevation, and Target Lesion Size on Treatment Outcomes With Lu-Dotatate: An Analysis of the NETTER-1 Study. European Journal of Nuclear Medicine and Molecular Imaging. 2020. Strosberg J, Kunz PL, Hendifar A, et al.

Cost-Effectiveness of [177Lu]Lu-Dotatate for the Treatment of Newly Diagnosed Advanced Gastroenteropancreatic Neuroendocrine Tumors: An Analysis Based on Results of the NETTER-2 Trial. Journal of Nuclear Medicine : Official Publication, Society of Nuclear Medicine. 2025. Holzgreve A, Unterrainer LM, Tiling M, et al.

Progression-free survival with Lutathera was 22.8mo (vs 8.5mo)

⇒ Lutathera is indicated for first-line therapy of higher grade neuroendocrine tumors

(singh2807?)